flucut

INFLUENZA A AMPLICON QC, BEFORE IRMA

Primers off, the internal-priming artefact caught.

flucut removes the universal RT-PCR primers (Uni12/Uni13) from influenza A reads and, with references, finds and measures **internal priming** — the hidden artefact that writes wrong bases into the consensus (the HA I348K lysine). One command before IRMA, no configuration, on Nanopore and Illumina.

1.6 s
for 32,000 reads
4 threads, libdeflate

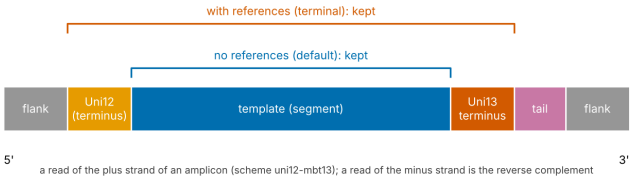
60 MB
of memory
runs on any laptop

99.6%
internally primed reads tolerated, still no N in the consensus
~91% for the IRMA-core trimmer

8 / 8
segments classified, terminal vs internal, by strand
reported, not trimmed silently

HOW IT WORKS

One read, one amplicon



a read of the plus strand of an amplicon (scheme uni12-mbt13); a read of the minus strand is the reverse complement

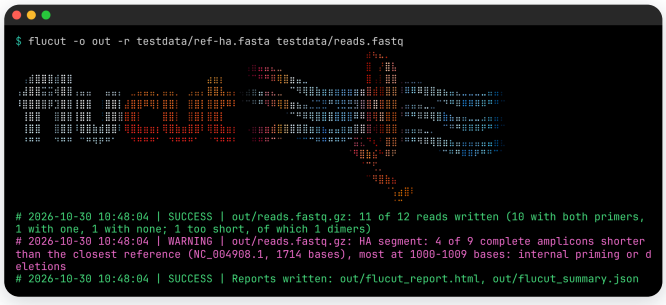
- the primer is searched only near the read end, aligned with errors, on both strands, degenerate bases and all
- the sequence after the primer (a probe) is located in the segment: at the terminus it is a real amplicon, deeper inside it is internally primed
- an internal or unsure end loses its **whole** primer (cut when unsure), so the artefact never reaches the consensus
- Nanopore adapters (native, rapid, ligation) removed at broken ends; one IRMA-ready FASTQ per sample

BUILT FOR THE PROTOCOL

No primer files, no tuning

Uni12 / Uni13 · mbt tailed · uni · Nanopore · Illumina · FASTQ / BAM · MinKNOW dirs

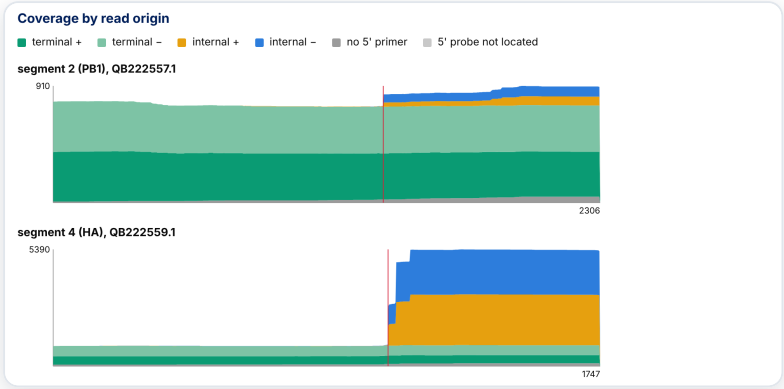
- the schemes and primers are built in — no `-g/-a/-e` to guess
- the error rate is measured on the first reads (`-e auto`); a Nanopore and an Illumina profile (`-p`)
- with `-r` , the conserved 12/13-base termini are kept at the ends confidently terminal — complete segment ends in one pass



WHAT THE OTHERS MISS

See the internal priming — don't just trim it

Every primer-aware trimmer removes the Uni12 footprint. Only flucut tells you it was there: it reconstructs the coverage of each segment from its own read classification, so internal priming shows as a step up above the full-length baseline — here the HA I348K site, in a real H5N1 Nanopore run.



Coverage by read origin (SRA SRR36760543, complete H5N1 segment references). Green: full-length amplicons, a flat baseline. Amber + blue: internally primed reads, by strand, beginning at the red site line. HA steps up sharply (the I348K artefact); PB1 shows a smaller internal site.

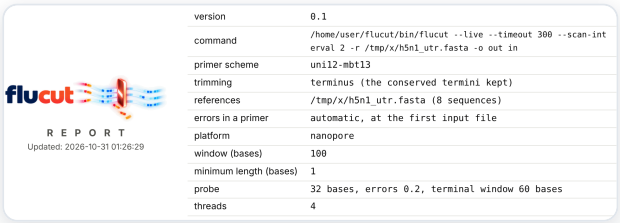
	flucut	cutadapt	BBDuk	IRMA-core	dorado	Barbell
Nanopore	✓	✓	✓	✓	✓	✓
Illumina	✓	✓	✓	✓	–	–
Max internal priming, no N in consensus†	99.6%	99.6%‡	93%	91%	99.7%	–
Keeps the correct segment ends in the consensus	✓	–	–	–	–	–
Internal-priming analysis + coverage plot	✓	–	–	–	–	–
Built-in flu protocol, no primer file	✓	–	–	✓	–	–
Speed / memory, 32k reads	1.6 s / 60 MB	3.7 s / 41 MB	4.0 s / 950 MB	7.5 s / 11 MB	8.1 s / 480 MB	2.4 s / 44 MB

† the highest share of internally primed reads at the HA site at which IRMA still calls the correct base; derived from the measured read retention (`benchmark/COMPARISON.md`). ‡ cutadapt reaches it only by also discarding ~27% of the genuine full-length reads. Barbell removes barcodes, not the internal Uni primer, so it cannot correct the artefact.

REAL TIME

Live, while you sequence

With `--live`, flucut follows a MinKNOW run: each new file is trimmed as it is written, and the report and JSON summary are updated and reloaded in the browser — the internal priming of the run building up live, on the sequencing laptop. Resume, interruption and end-of-run are handled, and the output matches a normal run on the final files.



GET FLUCUT

Free and open source

github.com/baia-ipc/flucut

C, ISC license, Linux and macOS, no dependencies.

```
$ make
$ flucut -r refs.fasta -o out/ \
   run/fastq_pass/
$ IRMA FLU-minion out/*.fastq.gz sample
```